



Comparative Analysis of Foundational Generative Models: DCGAN, WGAN-GP, and VAE

CSE 429: Computer Vision and Pattern Recognition

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1. Introduction

Deep generative models have revolutionized representation learning and content synthesis. Among the most impactful advances are:

- Generative Adversarial Networks (GANs) [?]
- Wasserstein GANs (WGANs) and their stabilized variants
- Variational Autoencoders (VAEs)

This report investigates three seminal papers:

- DCGAN: introducing stable convolutional GAN training
- WGAN-GP: resolving instability in adversarial training
- VAE: probabilistic autoencoders for variational inference

2. Paper 1: Unsupervised Representation Learning with Deep Convolutional GANs (DCGAN)

Radford, Metz & Chintala (2015)

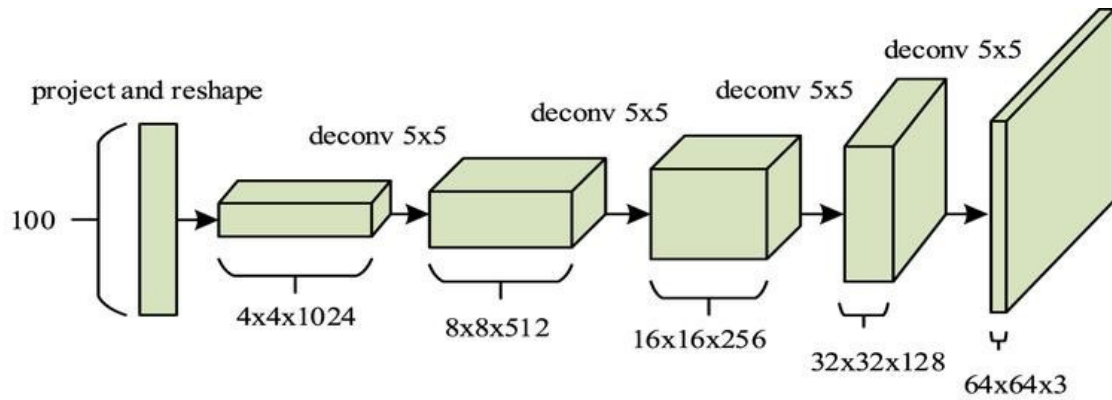
2.1 2.1 Motivation

DCGAN addresses a major challenge in GAN training: instability and sensitivity to architectural and optimization choices. The authors hypothesize that convolutional architectures, long successful in supervised vision tasks, can enhance GAN training if appropriately constrained.

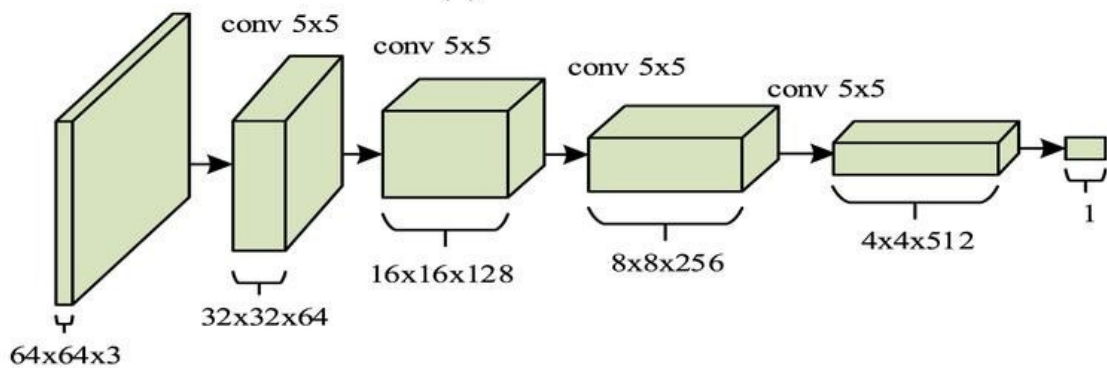
2.2 2.2 Architectural Constraints

The authors propose the DCGAN architecture with several crucial modifications:

- Remove fully connected layers in favor of deep convolutional layers
- Use strided convolutions (generator) and fractional-strided convolutions (discriminator)
- Batch normalization to stabilize gradients
- Tanh activation in the output layer of the generator for bounded pixel outputs
- Use LeakyReLU in the discriminator, ReLU in the generator



(a) Generator



(b) Discriminator

Figure 1: DCGAN architecture

2.3 2.3 Experiments and Results

Datasets: LSUN Bedrooms, CelebA Faces, ImageNet subsets

Results:

- Generator produced high-resolution, diverse images
- Latent space showed semantic interpolation
- Discriminator features repurposed for supervised classification (82.8% on CIFAR-10)
- Vector arithmetic in latent space (e.g., “man with glasses” - “glasses” + “woman”)

2.4 2.4 Strengths and Limitations

Strengths:

- Simple yet powerful design rules
- Robust unsupervised feature learning

- First GAN to show interpretable latent structure

Limitations:

- Still prone to mode collapse
- No theoretical justification for design choices
- Difficult to scale to very deep models

3. Paper 2: Improved Training of Wasserstein GANs (WGAN-GP)

Gulrajani et al. (2017)

3.1 Motivation and Background

Standard GANs optimize a loss based on Jensen-Shannon divergence, which can lead to:

- Vanishing gradients when distributions do not overlap
- Mode collapse (lack of diversity)
- Non-informative generator loss

WGAN reformulates the GAN objective using the **Wasserstein distance**:

$$W(P_r, P_g) = \inf_{\gamma \in \Pi(P_r, P_g)} E_{(x,y) \sim \gamma} [\|x - y\|]$$

Explanation: This expression defines the optimal cost of transporting the "mass" of generated distribution P_g to match the real distribution P_r . The infimum is over all joint distributions γ with marginals P_r and P_g . Unlike JS divergence, the Wasserstein distance provides a meaningful gradient even when distributions are disjoint.

3.2 Gradient Penalty Strategy

WGAN-GP introduces a gradient penalty to enforce the Lipschitz constraint:

$$E_{\hat{x} \sim P_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\|_2 - 1)^2]$$

Explanation: Rather than clipping weights (as in WGAN), WGAN-GP penalizes the critic's gradient norm from deviating from 1, ensuring smoother optimization and improved sample quality. This term regularizes the critic to remain close to a 1-Lipschitz function.

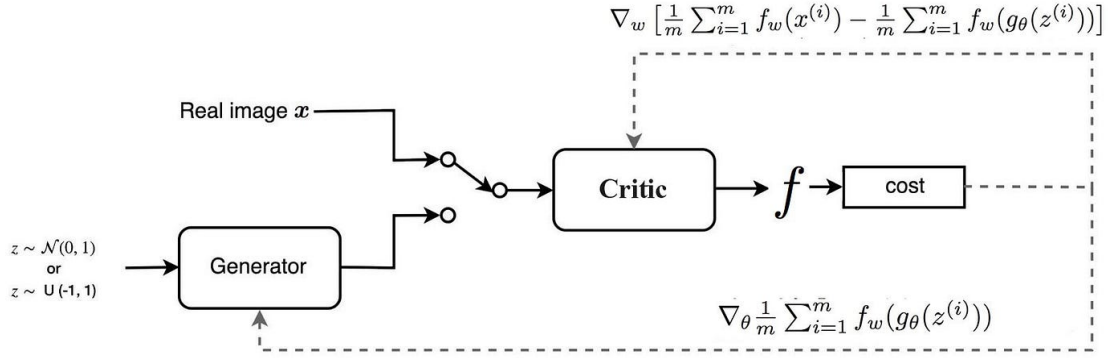


Figure 2: WGAN-GP architecture

3.3 3.3 Experimental Evaluation

Datasets: CIFAR-10, LSUN Bedrooms, 32×32 ImageNet

Results:

- Stable training with deep architectures (e.g., ResNet-101)
- Smooth convergence and high-quality samples
- Inception Score: 7.86 (unsupervised CIFAR-10)

3.4 3.4 Strengths and Limitations

Strengths:

- Alleviates mode collapse
- Provides meaningful loss metric
- Compatible with deep and complex architectures

Limitations:

- Gradient computation adds overhead
- Still requires hyperparameter tuning (e.g., λ penalty)

4. Paper 3: Auto-Encoding Variational Bayes (VAE)

Kingma & Welling (2013)

4.1 Probabilistic Formulation

VAE combines variational inference with autoencoding. The generative model is:

$$p_{\theta}(x, z) = p(z)p_{\theta}(x|z)$$

We optimize the Evidence Lower Bound (ELBO):

$$\log p(x) \geq E_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\phi}(z|x)||p(z))$$

Explanation:

- $q_{\phi}(z|x)$ is the encoder (approximate posterior)
- $p_{\theta}(x|z)$ is the decoder (likelihood)
- The first term is the expected reconstruction log-likelihood
- The second term is a regularizer (KL divergence) ensuring $q_{\phi}(z|x)$ stays close to a prior $p(z)$ (usually $\mathcal{N}(0, I)$)

4.2 Reparameterization Trick

To backpropagate through the sampling operation, VAE uses:

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

Explanation: This trick expresses the stochastic latent variable z as a deterministic function of the mean $\mu(x)$ and standard deviation $\sigma(x)$ plus noise. This enables the use of gradient-based optimization on stochastic layers.

4.3 AEVB Algorithm

The Auto-Encoding Variational Bayes (AEVB) algorithm jointly trains the encoder and decoder networks by optimizing the Evidence Lower Bound (ELBO). Key points include:

- The encoder $q_{\phi}(z|x)$ and decoder $p_{\theta}(x|z)$ are parameterized neural networks.
- The training objective is to maximize the ELBO:

$$E_{q_{\phi}(z|x)}[\log p_{\theta}(x|z)] - D_{\text{KL}}(q_{\phi}(z|x)||p(z))$$

- The **reparameterization trick** expresses the sampling operation as a differentiable transformation:

$$z = \mu(x) + \sigma(x) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

allowing gradients to propagate through stochastic nodes.

- **Stochastic Gradient Descent (SGD)** is employed to optimize the model parameters ϕ and θ . The term “stochastic” arises from:
 - Using mini-batches of data to estimate gradients, which introduces noise in gradient estimates.
 - Monte Carlo sampling from the latent distribution $q_\phi(z|x)$ to approximate expectations.
- The algorithm proceeds iteratively:
 1. Sample a mini-batch of input data x .
 2. Compute encoder outputs $\mu(x)$ and $\sigma(x)$.
 3. Sample latent variables z via the reparameterization trick.
 4. Generate reconstructions $\hat{x}$ through the decoder $p_\theta(x|z)$.
 5. Compute the ELBO loss (negative of the objective above).
 6. Compute gradients via backpropagation and update ϕ, θ using SGD or its variants (e.g., Adam).

This approach allows the VAE to efficiently optimize a probabilistic model with continuous latent variables while leveraging deep learning frameworks.

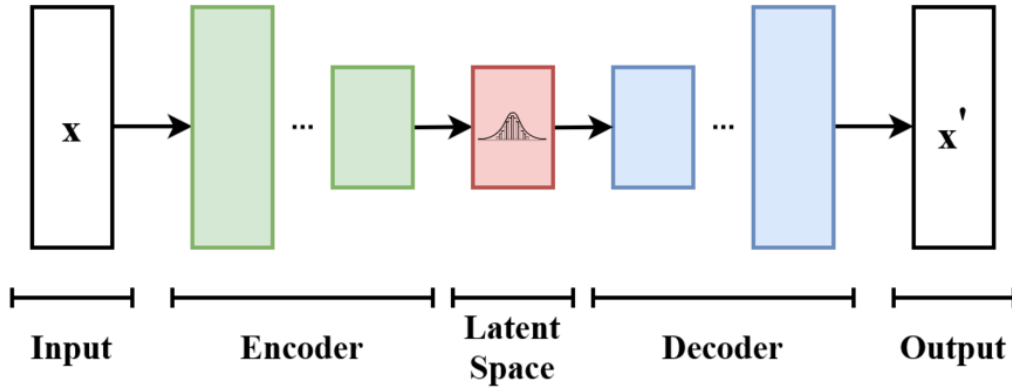


Figure 3: VAE architecture illustrating the encoder, reparameterization trick, and decoder.

4.4 Experimental Setup and Results

Datasets: MNIST, CIFAR-10

Results:

- Generated samples with diverse and coherent structure
- Learned latent space organized semantically
- Demonstrated robustness to noisy data

4.5 4.5 Strengths and Limitations

Strengths:

- Principled probabilistic framework
- Enables inference and generation in the same model
- Continuous latent space suitable for interpolation

Limitations:

- Generated samples tend to be blurrier than GANs
- KL term can cause latent space collapse
- Strong assumptions on posterior and prior distributions

5. Comparative Analysis

This section summarizes and contrasts the key characteristics of DCGAN, WGAN-GP, and VAE.

Aspect	DCGAN	WGAN-GP	VAE
Model Type	GAN with convolutional architectures	GAN with Wasserstein loss and gradient penalty	Probabilistic latent variable model (autoencoder)
Training Stability	Moderate; prone to mode collapse and instability	Improved stability via Wasserstein loss and gradient penalty	Stable due to explicit ELBO optimization
Loss Function	Jensen-Shannon divergence via adversarial loss	Wasserstein distance with gradient penalty	Evidence Lower Bound (ELBO) combining reconstruction and KL divergence
Output Quality	Sharp, realistic images but occasional mode collapse	High-quality samples with reduced mode collapse	Typically blurrier but more diverse samples
Latent Space	Learned implicitly; shows semantic structure	Learned implicitly; better continuity and stability	Explicit probabilistic latent variables with continuous distributions
Interpretability	Some interpretability in latent vectors (e.g., vector arithmetic)	Improved interpretability due to Wasserstein distance metric	Explicit probabilistic interpretation and inference capabilities
Computational Cost	Relatively low, standard GAN training	Higher due to gradient penalty calculation	Moderate; requires reparameterization but no adversarial training
Applications	Image synthesis, representation learning	High-quality image synthesis, stable GAN training	Generative modeling, representation learning, semi-supervised learning
Key Limitations	Mode collapse, training instability	Extra computation, hyperparameter sensitivity	Blurry outputs, assumptions on latent distributions

Table 1: Comparison of DCGAN, WGAN-GP, and VAE

6. Conclusion

This report analyzed three fundamental generative models: DCGAN, WGAN-GP, and VAE. DCGAN pioneered stable GAN training using convolutional architectures. WGAN-GP enhanced GAN stability and sample quality by reformulating the loss with Wasserstein distance and gradient penalties. VAE introduced a probabilistic approach to generative modeling using variational inference, balancing reconstruction and latent regularization.

Each model has unique strengths and limitations, making them suitable for different applications. Future research aims to combine advantages of GANs and VAEs for improved generative performance.

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